13 GASES IN THE ATMOSPHERE

This chapter looks at the most common gases in the atmosphere, concentrating on oxygen. We will also look at carbon dioxide, which is a greenhouse gas and may contribute to climate change.



▶ Figure 13.1 Despite the gases added by industry, the air around us is mostly nitrogen and oxygen.

LEARNING OBJECTIVES

- Know the approximate percentages by volume of the four most abundant gases in dry air
- Understand how to determine the percentage by volume of oxygen in air using experiments involving the reactions of metals (e.g. iron) and non-metals (e.g. phosphorus) with air
- Describe the combustion of elements in oxygen, including magnesium, hydrogen and sulfur
- Describe the formation of carbon dioxide from the thermal decomposition of metal carbonates, including copper(II) carbonate
- Know that carbon dioxide is a greenhouse gas and that increasing amounts in the atmosphere may contribute to climate change
- Practical: Determine the approximate percentage by volume of oxygen in air using a metal or a non-metal

THE COMPOSITION OF THE AIR

The approximate percentages (by volume) of the four most abundant gases present in *unpolluted, dry* air are shown in Table 13.1.

Table 13.1 Approximate percentages (by volume) of the main gases in unpolluted, dry air

Gas	Amount in air (%)	Amount in air (fraction)
nitrogen	78.1	about 4/5
oxygen	21.0	about 1/5
argon	0.9	
carbon dioxide	0.04	

There are also very small amounts of the other noble gases in the air.

KEY POINT

It is important to realise that these figures apply only to dry, unpolluted air. Air can have anywhere between 0 and 4% water vapour. The percentage of carbon dioxide in the air, although very small, is rising steadily because of human activity.

SHOWING THAT AIR CONTAINS ABOUT ONE-FIFTH OXYGEN

We will look at a few different methods for measuring the percentage of oxygen in the air. All the methods rely on the same basic principle: we react something with the oxygen in the air and look at how much the volume decreases as the oxygen is removed.

Safety Note: Wear eye protection and heat carefully to avoid cracking the glass syringe with the flame. Take care not to push the opposite plunger completely out!.

KEY POINT

When doing this experiment we need to make sure that we use enough copper to react with *all* the oxygen in the air - the copper must be in excess. There will be some unreacted copper metal at the end. If we do not use enough copper then the value we will get for the percentage of oxygen will be too low because not all of the oxygen will have been removed.

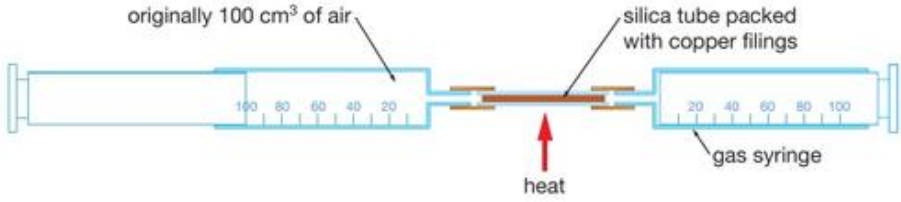
KEY POINT

We have put inverted commas around 'air' because we are now talking about air from which the oxygen has been removed, so it is mostly nitrogen. Nitrogen is very unreactive and so if you put a lighted splint into this 'air' it would be extinguished. Oxygen is needed for things to burn.

ACTIVITY 1

▼ PRACTICAL: USING COPPER

The apparatus shown in Figure 13.2 can be used to find the percentage of oxygen in the air.



▲ Figure 13.2 Using copper to measure the percentage of oxygen in air.

- The plunger on one of the gas syringes is pushed all the way in and the other moved out to 100 cm^3. We now know that the apparatus contains 100 cm^3 air.
- The silica tube is heated strongly (roaring Bunsen flame).
- The plunger in the left-hand gas syringe is pushed in. This causes the air to pass over the heated copper. This pushes out the plunger on the right-hand gas syringe.
- The plungers are pushed in sequence so that the air in the system keeps passing over the heated copper. The pink-brown copper turns black as copper(II) oxide is formed.
- As the copper reacts, the Bunsen burner is moved along the tube so that it is always heating fresh copper.
- The volume of gas in the syringes falls as the oxygen is consumed.
- We keep pushing the plungers in and out until there is no change in volume.
- The apparatus is then allowed to cool to room temperature again before taking the final volume of gas (because gases expand as they are heated).

RESULTS

initial volume of air in the system/cm^3	100
final volume of 'air' in the system/cm^3	79

The volume of the air has decreased because the oxygen has been removed as it reacts with the copper. The reaction that occurs is

$$2Cu(s) + O_2(g) \rightarrow 2CuO(s)$$

copper + oxygen → copper(II) oxide

We can work out from these data that the volume of oxygen that reacted was 100 – 79 = 21 cm^3.

The original volume of air was 100 cm^3 so we can work out the percentage of oxygen in the air:

percentage oxygen = 21/100 × 100 = 21%

KEY POINT

Iron needs oxygen and water to rust. The rusting of iron is discussed in Chapter 14.

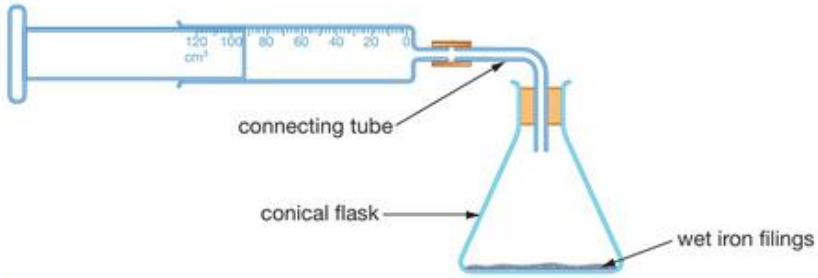
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Safety Note: Iron filings can irritate the skin and are a particular hazard if they get in the eyes.

ACTIVITY 2

▼ PRACTICAL: USING THE RUSTING OF IRON

Iron rusts in damp air, using up oxygen as it does so. We can use this reaction to determine how much oxygen there is in the air.



▲ Figure 13.3 This apparatus can be used to find the percentage of oxygen in the air.

We will use the apparatus from Figure 13.3, but before we start we need to know the volume of air present in the apparatus. We can find this by filling up the conical flask and connecting tube with water and then transferring the water to a measuring cylinder. On the conical flask we mark the position of the bung and only fill with water to that point. We will assume that the small volume occupied by the iron filings is negligible (very small compared to the overall volume).

The procedure for the experiment is as follows:

- Set up the apparatus as shown in Figure 13.3.
- Put wet iron filings into the conical flask.
- Record the initial reading on the gas syringe.
- Leave the apparatus in place for about a week, until the reading on the gas syringe stops changing.
- Record the final reading on the gas syringe.

volume of air in conical flask/cm^3	130
volume of air in connecting tube/cm^3	12
initial reading on gas syringe/cm^3	92
final reading on gas syringe/cm^3	43

So, the total volume of air inside the apparatus at the beginning of the experiment is 130 + 12 + 92 = 234 cm^3.

The total volume of 'air' in the apparatus at the end = 130 + 12 + 43
= 185 cm^3

Volume of oxygen used up = 234 – 185 = 49 cm^3

The percentage of oxygen in the air is 49/234 × 100 = 21%.

Sometimes when you do this experiment, the answer comes out as less than 21%. Possible reasons for this could be:

- The experiment was not left set up for long enough. The iron has not had enough chance to react with all the oxygen in the apparatus.
- Not enough iron was added at the beginning. The iron must be in excess, that is, there must be enough iron to react with all the oxygen present.

KEY POINT

Phosphorus does not react with water.

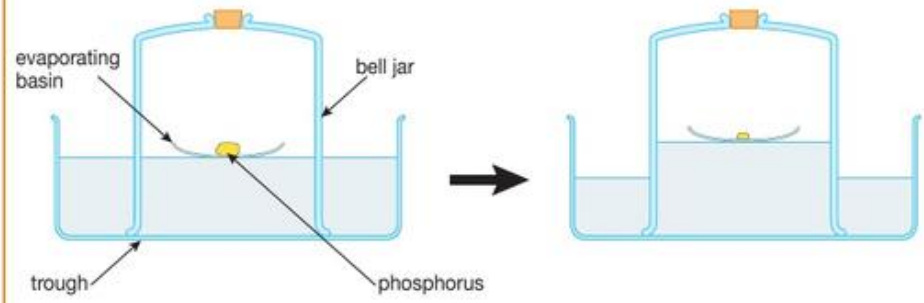
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Safety Note: The teacher demonstrating needs to wear goggles or a face shield, and have the room well ventilated. After igniting the phosphorous the bell-jar needs to be held down as pressure rises initially. After all the fumes have gone the evaporating basin needs to be sunk using the glass rod to prevent any unreacted phosphorous re-igniting.

ACTIVITY 3

▼ PRACTICAL: USING PHOSPHORUS

Phosphorus is a very reactive element that reacts with the oxygen in the air to form a phosphorus oxide. This oxide is very soluble in water.



▲ Figure 13.4 Finding the percentage of oxygen in air using phosphorus.

- The apparatus shown in Figure 13.4 is set up with the piece of phosphorus on an evaporating basin, which is floating in the water.
- The initial level of water is marked on the side of the bell jar with a waterproof pen or a sticker.
- The bung is removed from the bell jar and the phosphorus is touched with a hot metal wire in order to ignite it.
- The bung is quickly put back into the bell jar.
- The phosphorus burns, the bell jar becomes filled with a white smoke (phosphorus oxide) and the level of water rises inside the bell jar. The smoke eventually clears as the phosphorus oxide dissolves in the water.
- When the level of water inside the bell jar stops rising, the final level is marked.
- To find how much the water level has changed, the bell jar is turned upside down, filled with water to each mark in turn and the water is poured into a large measuring cylinder.

It is important that there is still some phosphorus left on the evaporating basin at the end of the experiment. We have used an excess of phosphorus so that there is more than enough to react with all the oxygen. If there was no phosphorus left, then we would probably get a lower value for the percentage of oxygen in the air because not all the oxygen might have been used up.

THE COMBUSTION OF ELEMENTS IN OXYGEN

Some elements burn in oxygen, these reactions are called *combustion reactions*. Elements burn more brightly and rapidly in pure oxygen than in air because air only contains 21% oxygen.

BURNING MAGNESIUM

Magnesium burns in oxygen with an extremely bright white flame to give a white, powdery ash of magnesium oxide:

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

The white powder formed is not very soluble in water but a very small amount does dissolve to form an alkaline solution:

$$MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(aq)$$



▲ Figure 13.5 Magnesium ribbon burning in air

BURNING SULFUR

Sulfur burns in oxygen with a blue flame. Poisonous, colourless sulfur dioxide gas is produced.

$$S(s) + O_2(g) \rightarrow SO_2(g)$$



▲ Figure 13.6 Sulfur burning in oxygen

KEY POINT

Sulfur dioxide is also called sulfur(IV) oxide.

The sulfur dioxide dissolves in water to form an acidic solution of *sulfurous acid*:

$$SO_2(g) + H_2O(l) \rightarrow H_2SO_3(aq)$$

KEY POINT

Sulfurous acid (H_2SO_3), not sulfuric acid (H_2SO_4), is formed.

BURNING HYDROGEN

Hydrogen burns in oxygen with a pale blue flame. The product is water:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

If you ignite a mixture of hydrogen and oxygen it will explode. This is the basis of the 'squeaky pop' test for hydrogen (see Chapter 18).

THE PROPERTIES OF OXIDES

We can make some generalisations about the properties of oxides formed when elements burn in oxygen.

METAL OXIDES

- Metal oxides are ionic compounds containing O^{2-} ions.
- Metal oxides are usually basic oxides, which means that they react with acids to form salts.
- Metal oxides are usually insoluble in water. Those metal oxides that are soluble in water react with it to form alkaline solutions containing hydroxide (OH^-) ions.

NON-METAL OXIDES

- Non-metal oxides are covalent compounds.
- Non-metal oxides are usually acidic oxides, which react with alkalis /bases to form salts.
- Non-metal oxides are often soluble in water and react with it to form acidic solutions containing hydrogen (H^+) ions.

CARBON DIOXIDE



▲ Figure 13.7 Copper(II) carbonate decomposes to form carbon dioxide when heated.

HINT

You should remember the colours of copper(II) carbonate and copper(II) oxide.

Carbon dioxide is a colourless gas that is most easily made in the laboratory by the reaction between dilute hydrochloric acid and calcium carbonate in the form of marble chips:

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

Carbon dioxide can also be obtained when metal carbonates are heated strongly.

Most carbonates split to give the metal oxide and carbon dioxide when you heat them. This is an example of **thermal decomposition**, breaking up something by heating it.

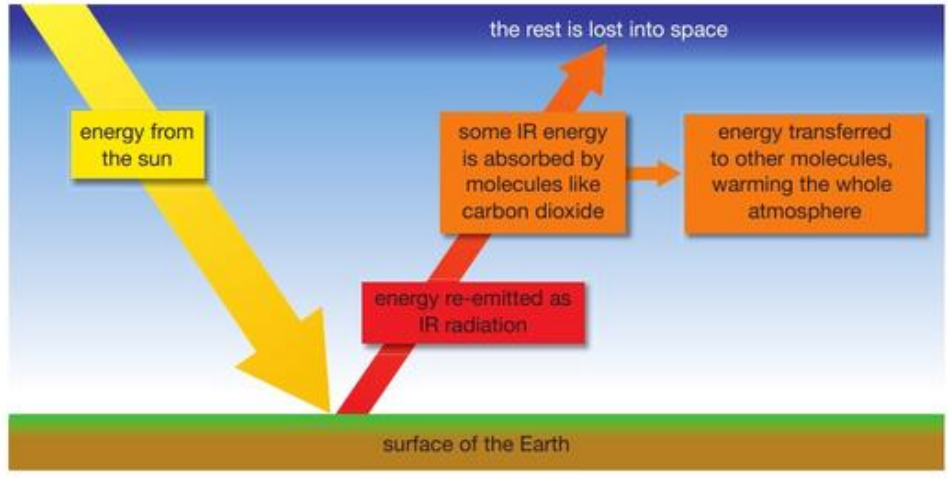
For example, copper(II) carbonate is a *green* powder which decomposes on heating to produce *black* copper(II) oxide:

$$CuCO_3(s) \rightarrow CuO(s) + CO_2(g)$$

Calcium carbonate doesn't decompose unless it is heated at quite high temperatures. This is a commercially important reaction because it is used to convert limestone (calcium carbonate) into quicklime (calcium oxide):

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

CARBON DIOXIDE AND GLOBAL WARMING: THE GREENHOUSE EFFECT



▲ Figure 13.8 The greenhouse effect

Carbon dioxide is produced when fossil fuels (coal, oil and gas) burn. For instance, when coal, which is mostly carbon, burns in excess oxygen:

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

REMINDER

A hydrocarbon is a molecule containing carbon and hydrogen only. These are discussed more fully in Unit 4.

Petrol is a mixture containing many different hydrocarbons. An example of a reaction that occurs when petrol burns is:

$$2C_8H_{18}(l) + 25O_2(g) \rightarrow 16CO_2(g) + 18H_2O(l)$$

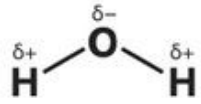
DID YOU KNOW?

Although most scientists believe that increases in carbon dioxide will cause devastating climate change, there are others who reject this. They claim that the increases in average world temperatures during the 20th century were part of natural heating and cooling cycles caused by the sun and changes to ocean currents. They believe the effect of carbon dioxide is small compared with these.

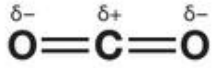
Carbon dioxide is a **greenhouse gas**. The greenhouse effect occurs when high-energy **UV** and visible light from the sun pass through the atmosphere and warm up the surface of the Earth. The surface of the Earth (like any other warm surface) radiates infrared (IR) radiation. This IR radiation is absorbed by molecules such as CO_2 in the atmosphere. These then give out this energy again in all directions, heating the atmosphere.

For approximately the last 200 years the level of CO_2 in the atmosphere has been increasing. This has occurred since the industrial revolution and is due to the burning of fossil fuels and deforestation (cutting down trees to create more land for agriculture). Many people believe that the increase in the level of carbon dioxide in the atmosphere may contribute to climate change. The exact nature of this climate change is difficult to know, but some of the things that scientists believe could happen are:

- polar ice caps could melt
- sea levels could rise
- there could be more extreme weather (such as floods, droughts and heat waves).



▲ Figure 13.9 Water is a polar molecule. The oxygen atoms attract the electrons in the covalent bonds more than the hydrogen atoms. The δ+ (delta positive) and δ– (delta negative) represent small charges on the atoms.



▲ Figure 13.10 The oxygen atoms attract the electrons in the covalent bonds more than the carbon atoms do (oxygen is more electronegative than carbon) but the charges cancel due to the shape (a better way of saying this is that the dipoles cancel).

LOOKING AHEAD

Why is CO_2 a greenhouse gas and oxygen and nitrogen are not?

The presence of greenhouse gases in the atmosphere is essential to the maintenance of life on Earth as they keep the planet at a temperature at which water is mostly a liquid. However, it is very important that we do not have too much of these greenhouse gases. If nitrogen and oxygen, the main constituents of the atmosphere, were greenhouse gases the temperature of the Earth would be too high to support life as we know it. So, what makes something a greenhouse gas? A greenhouse gas must be able to absorb IR radiation emitted by the surface of the Earth and then emit it again in all directions, warming the atmosphere. All molecules vibrate and if these vibrations involve a change in the polarity of the molecule then the molecule will be able to absorb IR radiation and be a greenhouse gas.

What is polarity? A molecule is polar if one end of the molecule is slightly negative and the other end is slightly positive. Polarity arises because different atoms attract the electrons in a covalent bond to different extents (this is a property called *electronegativity*). The polarity also depends on the shape of the molecule. Water is a good example of a polar molecule.

The polarity of a water molecule changes as it vibrates and water vapour is an important greenhouse gas.

Carbon dioxide is a non-polar molecule because it is linear (the atoms are in a straight line) and symmetrical, and therefore there is no positive end and no negative end of the molecule.

However, as a CO_2 molecule vibrates it becomes non-symmetrical and polar. Because the polarity changes as a CO_2 molecule vibrates it can absorb IR radiation and is a greenhouse gas.

O_2 and N_2 are non-polar because both the atoms are the same and therefore attract the electrons in the covalent bonds equally, so there are no small charges. The molecules can only vibrate in one way and the molecules remain non-polar as they vibrate. This means that there is no change in polarity as they vibrate, therefore N_2 and O_2 cannot absorb IR radiation and are not greenhouse gases.

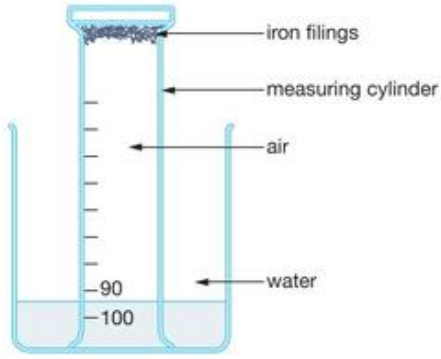
CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 5

1 State the approximate percentage of each of the following gases in dry air:
a nitrogen b oxygen c carbon dioxide d argon

SKILLS PROBLEM SOLVING 7

2 A student carried out an experiment to measure the percentage oxygen in the air by using the rusting of iron. The experimental set-up is shown below. The initial reading on the measuring cylinder was 95 cm^3. They left the apparatus for 2 days and the new reading on the measuring cylinder was 80 cm^3.



a Calculate the percentage oxygen in the air using this data.

SKILLS REASONING, DECISION MAKING

b Explain why the answer is not the same as you had expected and how the experiment could be improved.

SKILLS DECISION MAKING 6

3 A student carries out an experiment to investigate how much gas is produced when various metal carbonates are heated. Their data are shown in the table.

Metal carbonate	Mass of carbonate used/g	Volume of gas collected at rtp/cm^3	Mass of solid left at the end/g
lead(II) carbonate	1.00	89	0.84
copper(II) carbonate	1.00	194	0.64
sodium carbonate	1.00	0	1.00

a Describe an experiment that would allow the student to collect these data.

SKILLS CRITICAL THINKING 5

b Name the gas collected.

SKILLS PROBLEM SOLVING 9

c Write an equation for the reaction occurring with lead(II) carbonate.

SKILLS CRITICAL THINKING 5

d Name the type of reaction occurring with lead(II) carbonate and copper(II) carbonate.

SKILLS REASONING 6

e Explain what these results indicate about the thermal stability of sodium carbonate compared to the other two substances.

7

f Explain why much more gas is given off in the copper(II) carbonate experiment than in the lead(II) carbonate experiment.

SKILLS PROBLEM SOLVING

g Use the data from the lead(II) carbonate experiment to calculate the mass of 100 cm^3 of the gas.

SKILLS EXECUTIVE FUNCTION

h The student repeated the experiment for copper(II) carbonate and only obtained 152 cm^3 of gas this time. Explain two things that could have gone wrong with the second experiment to give this result.

SKILLS CRITICAL THINKING 6

4 Carbon dioxide is described as a *greenhouse gas*. Explain what this means.